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Binbin Liu

2627128848@qq.com

Anhui and Huaihe River Institute of Hydraulic Research

Mingming Wang

Anhui and Huaihe River Institute of Hydraulic Research

Xiaolei Zhu

Anhui and Huaihe River Institute of Hydraulic Research

Wanbo Zhang

Anhui and Huaihe River Institute of Hydraulic Research

Article

Keywords:

Posted Date: June 22nd, 2026

DOI: <https://doi.org/10.21203/rs.3.rs-9809809/v1>

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Additional Declarations: No competing interests reported.

A Health-Score-Based Framework for Quality Assessment and Spatial Reconstruction of Rainfall Monitoring Data in Reservoir Watersheds

Binbin Liu^{1,2}, Mingming Wang^{1,2}, Xiaolei Zhu^{1,2}, Wanbo Zhang^{1,2}

¹ Anhui and Huaihe River Institute of Hydraulic Research, Hefei 230088, China

² Anhui Provincial Key Laboratory of Water Science and Smart Water Conservancy, Hefei 230088, China

Correspondence: 2627128848@qq.com

Abstract

Reliable rainfall records are essential for flood forecasting, reservoir operation and water resource management, yet automatic rain-gauge networks often contain persistent fixed values, long data gaps, isolated spikes and short peaks without realistic recession behavior. We present a model-driven framework for diagnosing and reconstructing rainfall monitoring data from the 100-station Meishan Reservoir network in Anhui, China. The framework couples rule-based anomaly detection with a station Health Score that summarizes data quality and guides donor selection for inverse distance weighting reconstruction. Across the network, 21 stations were classified as poor or critical quality. In leave-one-out tests against arithmetic mean, linear interpolation, cubic spline interpolation and nearest-neighbour interpolation, the proposed approach achieved the best performance, with RMSE 0.281 mm, MAE 0.182 mm, NSE 0.979 and PBIAS -3.82%. RMSE was 58.9–90.8% lower than that of the benchmark methods. The framework is transparent, independent of database architecture and readily deployable in operational hydrological information systems for rainfall quality control and gap repair.

1 Introduction

Accurate, gap-free precipitation records are the primary input for hydrological models used in flood routing, drought assessment, and reservoir management [1, 2]. Ground-based rain gauge networks remain the most direct source of point precipitation data, providing temporal resolutions as fine as 5 minutes [3]. In operational networks, however, telemetric rain gauges are exposed to a spectrum of failure modes: mechanical clogging of tipping buckets produces unrealistic constant-drip readings; power or communication outages create extended data voids; electronic noise or debris contamination introduces isolated extreme spikes; and transient sensor glitches yield brief anomalous peaks without the temporal context of genuine rainfall events [4–6].

Left undetected and uncorrected, these anomalies propagate into hydrological model inputs, biasing flood peak estimates and reservoir inflow calculations. In China, where more than 90,000 reservoirs have been constructed and flood-related socioeconomic losses continue to rise under climate change [7, 8], the operational integrity of telemetric rainfall networks is directly tied to public safety. The Meishan Reservoir watershed in Anhui Province is an operational case: its 100-station rain gauge network supports real-time flood forecasting for a catchment receiving 60%–70% of annual precipitation during the East Asian Summer Monsoon [9, 10]. Traditional quality-control approaches — range checks, temporal difference tests, and spatial consistency checks — are applied as separate filters and focus on a single anomaly type at a time [3, 4, 11, 12]. No unified framework simultaneously: (i) identifies multiple co-occurring anomaly classes; (ii) translates the aggregate anomaly burden into a single, interpretable station quality index; and (iii) uses that quality index to filter donors for spatial reconstruction.

Recent research has begun to address individual aspects of this problem. Osrodka et al. [3] developed the RAINGAUGEQC scheme that assigns a quantitative quality index (QI) to individual telemetric rain gauge measurements in real time. El Hachem et al. [12] evaluated three open-source algorithms for quality control of personal weather station rainfall data and demonstrated that none universally outperforms the others, underscoring the need for multi-check frameworks. For missing data reconstruction, Golkhatmi and Farzandi [13] proposed a spatiotemporal quality control and MICE-based imputation approach for large precipitation datasets, while Hirca and Eryilmaz Turkkan [14] compared classical and machine-learning imputation methods for missing rainfall estimation. Spatial interpolation of precipitation remains an active area; Huang et al. [15] demonstrated that hybrid kriging–machine-learning approaches outperform purely geostatistical methods in complex terrain. For anomaly detection in temporal sensor data, Zamanzadeh Darban et al. [16] provide a comprehensive taxonomy of deep-learning methods. Despite these advances, no existing framework integrates anomaly detection, health scoring, and health-filtered spatial reconstruction into a single operational pipeline.

The present study addresses all three gaps with a framework derived from the operational software deployed at the Meishan Reservoir Management Bureau. The specific contributions are:

1. A four-category rule-based anomaly detection algorithm whose thresholds are physically motivated by the 0.5 mm tipping-bucket standard and a 45-day maintenance cycle.
2. A **Health Score** (HS) metric that penalises detected anomalies by type and severity, producing a scalar quality ranking directly actionable for maintenance prioritisation.
3. A health-score-filtered idw reconstruction pipeline that ensures only reliable donors contribute to spatial interpolation.
4. A rigorous comparative evaluation on the 100-station Meishan network with real two-year records, benchmarked against four alternative interpolation methods.

The remainder of the paper is organised as follows. Section 2 describes the study area and data. Section 3 presents the complete methodology. Section 4 reports the experimental results. Section 5 discusses limitations and future directions. Section 6 concludes.

2 Study Area and Data

2.1 Meishan Reservoir Watershed

The Meishan Reservoir is located in Jinzhai County, Lu'an City, Anhui Province, China (115°39'–115°56' E, 31°12'–31°44' N), situated in the upper Shi River (*Shi He*) sub-basin of the Huai River system (Figure 1). The reservoir controls a 1970 km² catchment and provides flood control, irrigation, and municipal water supply for downstream communities. The terrain is mountainous to hilly, with elevations ranging from 200 m to 1774 m a.s.l. The watershed experiences a humid subtropical monsoon climate; mean annual precipitation is 1380 mm, with 60%–70% falling between June and September during the East Asian Summer Monsoon [9, 10]. The pronounced wet–dry seasonality and complex orography create strong spatial variability in precipitation, making high-quality gauge data especially critical for operational flood forecasting [7].

2.2 Rain Gauge Network

The monitoring network comprises **100 automatic telemetry rain gauge stations** distributed across the watershed and surrounding administrative areas (Figure 1a). Station types include standard tipping-bucket gauges with a minimum measurable increment of 0.5 mm **per tip event** [6]. Observations are transmitted in

real time via GPRS to a central database and logged at sub-hourly intervals. Station elevations and inter-station spacings span wide ranges, with valley stations spaced 3 km to 5 km apart and remote mountain stations up to 15 km to 20 km from their nearest neighbours.

2.3 Dataset

The dataset used in this study covers a continuous **two-year** monitoring period (2023–2024), yielding up to 17 520 hourly records per station and a nominal total of 1 752 000 station-hours of raw observations. Data are queried from an operational relational database using SQL with the standard precipitation table and station metadata table. For this paper, the methodology is presented as a model framework independent of any specific database engine. The health-score results for all 100 stations are shown in Figure 1b; scores range from -95 to 100 (mean $\pm$ SD = 75.2 ± 28.4). The reservoir and its network form part of the national small-reservoir monitoring system that has undergone substantial quality improvement initiatives since 2021 [17].

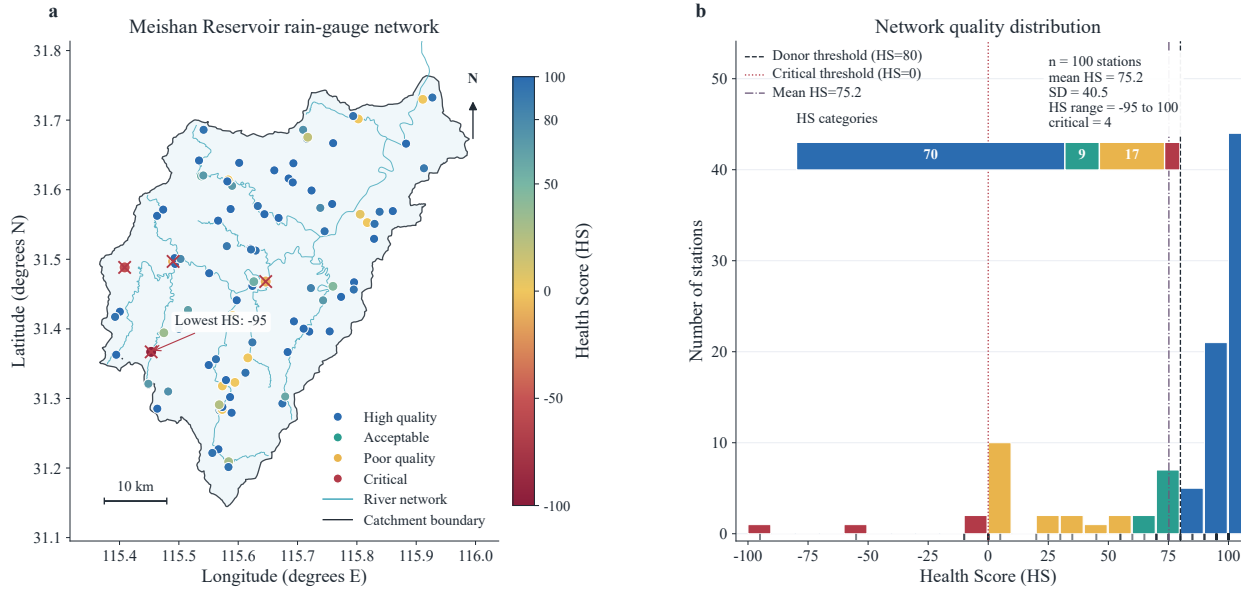


Figure 1: Meishan Reservoir rain gauge network. **(a)** Spatial distribution of 100 stations coloured by Health Score (HS); blue indicates high quality ($HS \geq 80$), red indicates critical quality ($HS < 0$); river network (blue lines) and watershed boundary are overlaid. **(b)** Frequency distribution of health scores across all 100 stations; vertical dashed lines mark the quality threshold ($HS = 80$) and the critical threshold ($HS = 0$).

3 Methodology

Figure 2 summarises the overall framework. Sections 3.1–3.6 describe each component in detail.

3.1 Notation and Preprocessing

Let $S = \{s_1, s_2, \dots, s_N\}$ denote the set of $N = 100$ monitoring stations and $\mathcal{T} = \{t_1, t_2, \dots, t_M\}$ the discrete observation time grid over the study period, where t_k is the k -th hourly timestamp. For station s_i , the raw

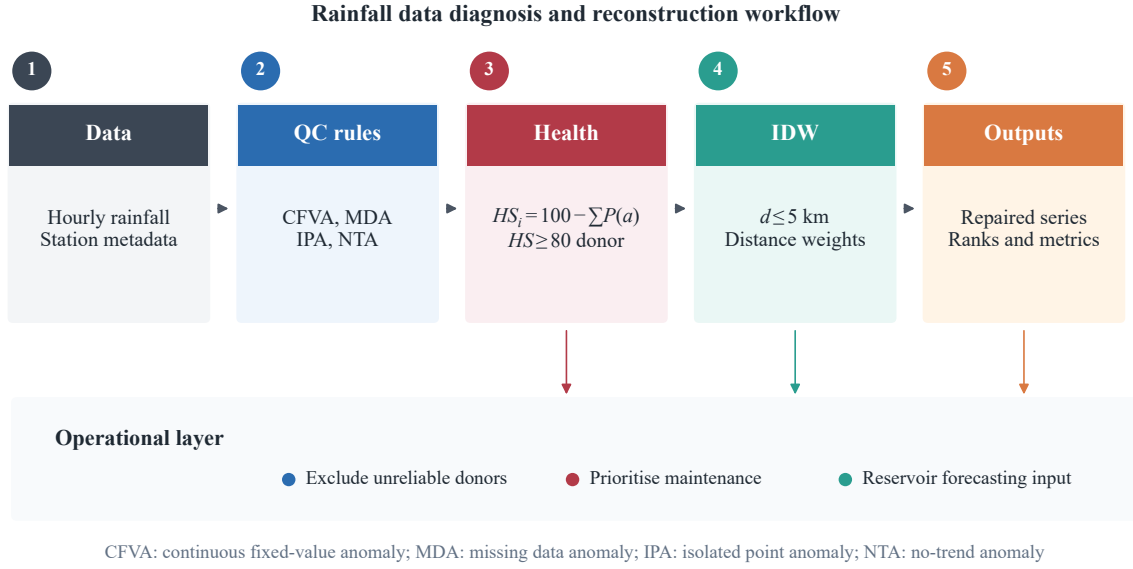


Figure 2: System architecture of the proposed health-score-driven rainfall quality diagnosis and reconstruction framework.

observation sequence is defined as:

$$\mathcal{R}_i = \{(t_k, r_{i,k}) \mid t_k \in \mathcal{T}, r_{i,k} \in \mathbb{R}_{\geq 0} \cup \{\text{NULL}\}\} \quad (1)$$

where $r_{i,k}$ (mm) is the recorded rainfall accumulation at station s_i during the interval ending at t_k , and NULL denotes a missing record.

Preprocessing comprises four steps applied in order:

(P1) Zero-value separation.

Records with $r_{i,k} = 0$ represent confirmed dry intervals and are excluded from anomaly analysis. Only non-zero, non-null records are passed to the detection algorithm, consistent with standard tipping-bucket gauge conventions [3, 4].

(P2) Type coercion.

The rainfall field is cast to double-precision floating point; non-numeric entries are set to NULL.

(P3) Deduplication.

Duplicate (s_i, t_k) pairs are resolved by retaining the record with the larger non-zero value.

(P4) Sorting.

The retained sequence is sorted ascending by t_k before detection begins.

Let $\mathcal{R}_i^+ = \{(t_k, r_{i,k}) \in \mathcal{R}_i \mid r_{i,k} > 0\}$ denote the preprocessed non-zero sequence for station s_i , with $n_i = |\mathcal{R}_i^+|$.

3.2 Multi-Type Anomaly Detection

The detection algorithm identifies four anomaly categories, applied sequentially to $\mathcal{R}_i^+$.

3.2.1 Continuous Fixed-Value Anomaly (CFVA)

A **Continuous Fixed-Value Anomaly** occurs when the tipping-bucket mechanism becomes mechanically stuck, reporting a constant drip rate rather than actual precipitation [6]. Such stuck sensors produce extended runs of values exactly equal to one tip (0.5 mm) or a small integer multiple thereof.

Definition. Let $\mathcal{V}_{\text{fixed}} = \{v \in \mathbb{R}^+ \mid v \in \{0.5, 1.0, 1.5\} \text{ mm}\}$ be the set of fixed-value indicators. A contiguous sub-sequence $\{r_{i,k}, r_{i,k+1}, \dots, r_{i,k+L-1}\} \subset \mathcal{R}_i^+$ is flagged as a CFVA if and only if:

$$r_{i,k+j} \in \mathcal{V}_{\text{fixed}} \quad \forall j \in \{0, 1, \dots, L-1\} \quad (2)$$

$$L \geq L_{\min} \quad (3)$$

where $L_{\min} = 50$ is the minimum run-length threshold (empirically set to prevent false positives during light drizzle while capturing sustained mechanical failures [6]). After detection, adjacent CFVA segments separated by fewer than 1 hour are merged into a single anomaly block before penalty computation (Section 3.3).

3.2.2 Missing Data Anomaly (MDA)

A **Missing Data Anomaly** is declared whenever the temporal gap between two consecutive non-zero observations (or between the study-period boundary and the nearest observation) exceeds a threshold $\Delta t_{\min}$.

Definition. Let $t_j^{(i)}$ and $t_{j+1}^{(i)}$ denote consecutive observation timestamps in $\mathcal{R}_i^+$. The inter-observation gap is:

$$\delta_j^{(i)} = t_{j+1}^{(i)} - t_j^{(i)} \quad (4)$$

The gap is classified as an MDA if:

$$\delta_j^{(i)} > \Delta t_{\min} \quad (5)$$

with $\Delta t_{\min} = 45$ days. This threshold reflects the maximum permissible outage duration in the Meishan network maintenance protocol: outages shorter than 45 days are treated as recoverable from surrounding valid records; outages exceeding 45 days correspond to documented hardware or communication failures [17]. Two boundary cases are handled separately:

- **Start-of-record MDA:** If $t_1^{(i)} - t_{\text{start}} > \Delta t_{\min}$, an MDA is recorded for the period $[t_{\text{start}}, t_1^{(i)}]$.
- **End-of-record MDA:** If $t_{\text{end}} - t_{n_i}^{(i)} > \Delta t_{\min}$, an MDA is recorded for the period $[t_{n_i}^{(i)}, t_{\text{end}}]$.

3.2.3 Isolated Point Anomaly (IPA)

An **Isolated Point Anomaly** is a single extreme observation that cannot be corroborated by temporally adjacent non-zero records, consistent with electronic noise or debris contamination rather than a genuine extreme precipitation event. Deep-learning-based anomaly detection literature defines this pattern as an “isolated point” anomaly in univariate time series, i.e., a value that deviates markedly from its neighbourhood without temporal context [16].

Definition. The long-term statistics of non-zero observations at station s_i are:

$$\mu_i = \frac{1}{n_i} \sum_{(t_k, r_{i,k}) \in \mathcal{R}_i^+} r_{i,k} \quad (6)$$

$$\sigma_i = \sqrt{\frac{1}{n_i} \sum_{(t_k, r_{i,k}) \in \mathcal{R}_i^+} (r_{i,k} - \mu_i)^2} \quad (7)$$

An observation $r_{i,k}$ satisfies the **extreme condition** if:

$$r_{i,k} > \mu_i + 3\sigma_i \quad \text{or} \quad r_{i,k} < \mu_i - 3\sigma_i \quad (8)$$

Given that the extreme condition is satisfied, let $\mathcal{W}(t_k) = \{(t_j, r_{i,j}) \in \mathcal{R}_i^+ \mid |t_j - t_k| \leq 5 \text{ h}, j \neq k\}$ be the set of non-zero neighbours within a $\pm 5 \text{ h}$ temporal window. The observation is classified as an **IPA** if:

$$|\mathcal{W}(t_k)| < 3 \quad (9)$$

indicating the extreme value is isolated in time and lacks physical corroboration.

3.2.4 No-Trend Anomaly (NTA)

A **No-Trend Anomaly** occurs when an extreme observation has sufficient temporal neighbours ($|\mathcal{W}(t_k)| \geq 3$), but the surrounding sequence does not exhibit the progressive intensification and gradual recession expected of a genuine high-intensity rainfall event in the East Asian monsoon context [18].

Definition. Given the extreme condition (8) and $|\mathcal{W}(t_k)| \geq 3$, let $r_{i,k-2}^{(+)}$, $r_{i,k-1}^{(+)}$, $r_{i,k}^{(+)}$, $r_{i,k+1}^{(+)}$ denote the two preceding, the current, and the first following non-zero observations. The observation is classified as an **NTA** if:

$$r_{i,k-2}^{(+)} < r_{i,k-1}^{(+)} \quad \text{and} \quad r_{i,k+1}^{(+)} < r_{i,k-1}^{(+)} \quad (10)$$

Equation (10) captures the pattern: two observations increase before the extreme value, but the next observation drops below the pre-peak level — inconsistent with the gradual recession of a real convective or frontal event in the Meishan monsoon regime [18].

Figure 3 illustrates all four anomaly types.

3.3 Station Health Scoring

To summarise the anomaly burden of each station into a single operational quality index, we define the **Health Score** (HS) as a 100-point deduction-based metric analogous to the quantitative quality index (QI) concept applied in operational rain gauge networks [3]:

$$\text{HS}_i = 100 - \sum_{a \in \mathcal{A}_i} P(a) \quad (11)$$

where $\mathcal{A}_i$ is the set of detected anomaly instances at station s_i , and $P(a) \geq 0$ is the penalty (in points) associated with anomaly instance a . The penalty schedule is summarised in Supplementary Table S1.

where D_a is the duration of a merged CFVA block in days. End-of-record MDAS receive the highest penalty (100 points, which can drive HS below zero) because they imply the station has completely ceased reporting, rendering its recent data entirely unavailable for operational use.

Supplementary Table S2 lists the operational quality categories based on HS.

Figure 4 shows the penalty schedule and decomposed health scores for 12 selected stations.

3.4 Spatial Reconstruction via IDW Interpolation

Stations with $\text{HS} < \text{HS}_{\min} = 80$ are designated as **target stations** (S_L). Stations with $\text{HS} \geq 80$ form the **donor pool** (S_H).

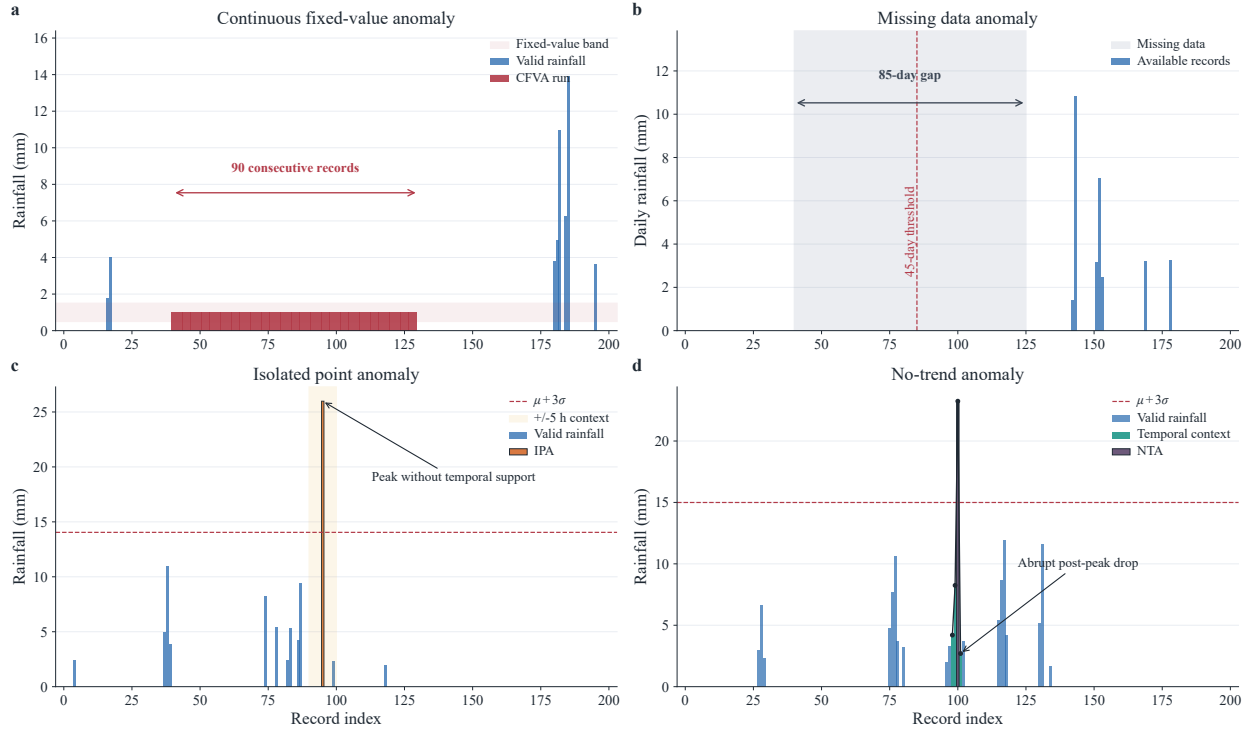


Figure 3: Examples of the four anomaly types detected by the proposed algorithm. **(a)** CFVA: 90 consecutive records at 1.0 mm (≥ 50 -record threshold) indicating a stuck tipping-bucket. **(b)** MDA: an 85-day data void exceeding the 45-day threshold. **(c)** IPA: a single-hour extreme value ($> \mu + 3\sigma$) with fewer than 3 non-zero neighbours in the ± 5 h window. **(d)** NTA: an extreme peak with sufficient temporal context but lacking the expected gradual post-peak recession.

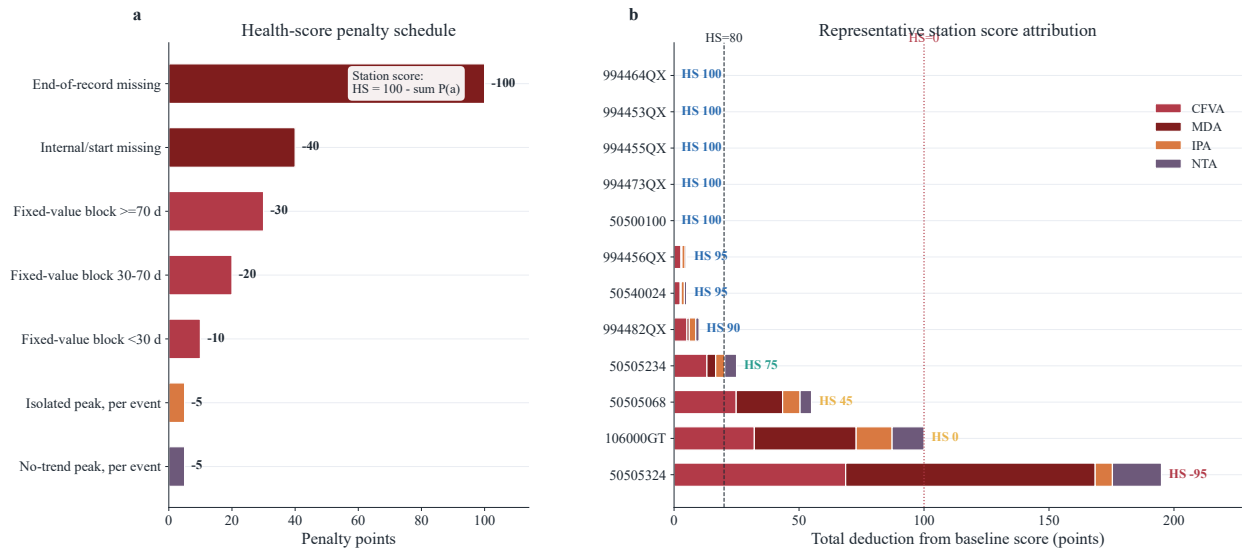


Figure 4: Health Score system. **(a)** Penalty schedule by anomaly type and severity. **(b)** Decomposition of health scores for 12 selected stations (S1–S12) sorted by final HS; stacked bars show the contribution of each anomaly type to the total deduction from the base score of 100.

3.4.1 Haversine Distance

The great-circle distance between stations s_i and s_j with WGS-84 geographic coordinates (ϕ_i, λ_i) and (ϕ_j, λ_j) (radians) is:

$$d_{ij} = 2R_{\oplus} \arcsin \left(\sqrt{\sin^2 \frac{\phi_j - \phi_i}{2} + \cos \phi_i \cos \phi_j \sin^2 \frac{\lambda_j - \lambda_i}{2}} \right) \quad (12)$$

where $R_{\oplus} = 6371$ km is the mean Earth radius. Equation (12) provides accurate inter-station distances for the mountainous terrain of the Meishan watershed, where planar approximations would introduce non-trivial errors at inter-station separations of 5–20 km.

3.4.2 Neighbour Selection

For a target station $s_{\text{tgt}} \in \mathcal{S}_L$, the **active neighbour set** is:

$$\mathcal{N}(s_{\text{tgt}}) = \{s_j \in \mathcal{S}_H \mid d_{\text{tgt},j} \leq d_0\} \quad (13)$$

where $d_0 = 5$ km is the search radius. The health-score filter $s_j \in \mathcal{S}_H$ ($\text{HS} \geq 80$) ensures that only reliable donors contribute, preventing error propagation from degraded stations [11, 12]. In sparse mountain zones where $|\mathcal{N}(s_{\text{tgt}})| = 0$ at $d_0 = 5$ km, the radius is adaptively expanded in 5 km increments up to a maximum of 40 km.

Figure 5 illustrates the neighbour selection geometry.

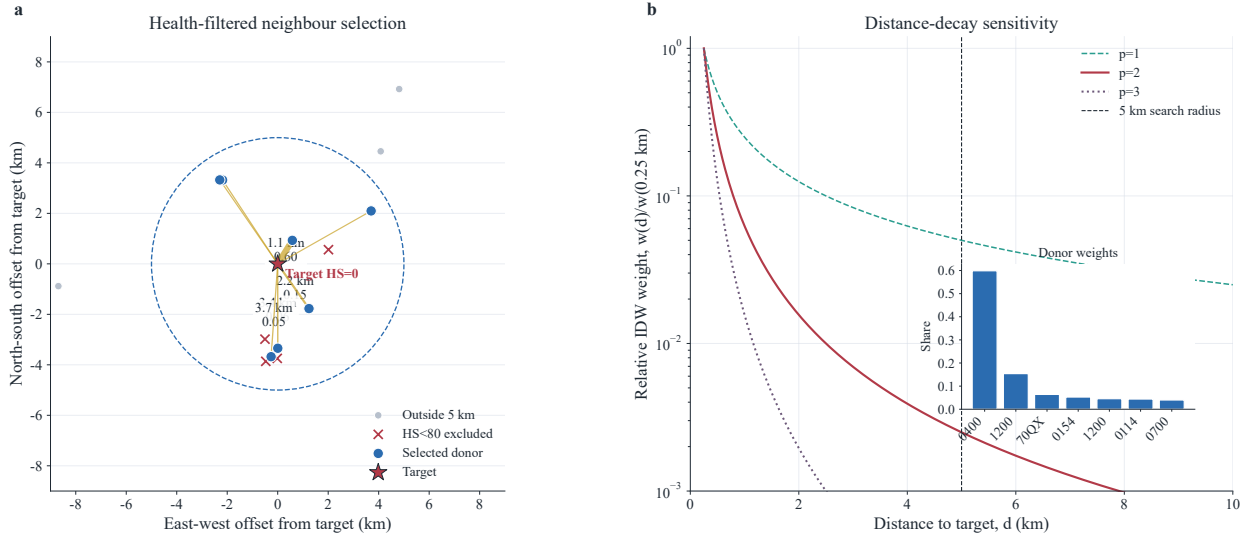


Figure 5: IDW spatial reconstruction. **(a)** Neighbour station selection for a selected target station (★); selected donors (blue circles, $\text{HS} \geq 80$, $d \leq 5$ km) are connected to the target with weighted links; stations outside the radius (grey) or below the health threshold (red ×) are excluded. **(b)** Normalised IDW weight as a function of distance for three power-parameter values; $p = 2$ (solid red, used in this study) provides balanced distance decay.

3.4.3 IDW Interpolation

For each time step t_k within a missing or anomalous segment at station s_{tgt} , the **reconstructed rainfall** $\hat{r}_{\text{tgt},k}$ is estimated as the distance-weighted average of concurrent donor observations [1, 2]:

$$\hat{r}_{\text{tgt},k} = \frac{\sum_{s_j \in \mathcal{N}(s_{\text{tgt}})} w_j(d_{\text{tgt},j}) r_{j,k}}{\sum_{s_j \in \mathcal{N}(s_{\text{tgt}})} w_j(d_{\text{tgt},j})} \quad (14)$$

with the **idw weight**:

$$w_j(d_{\text{tgt},j}) = \frac{1}{d_{\text{tgt},j}^p + \varepsilon} \quad (15)$$

where $p = 2$ and $\varepsilon = 10^{-6} \text{ km}^2$ (preventing division by zero when a donor coincides with the target). The power $p = 2$ is the standard choice for precipitation interpolation, providing a distance-decay profile that balances the contribution of near and intermediate donors [2, 19]. If donor s_j has a missing record at time step t_k , that donor is excluded from both the numerator and denominator of equation (14) for that time step only.

3.5 Benchmark Interpolation Methods

Four benchmark methods are evaluated under the same $\mathcal{N}(s_{\text{tgt}})$: the set spans spatial averaging, temporal interpolation, spline interpolation, and nearest-neighbour resampling, matching method families used in recent rainfall and hydrological gap-filling comparisons [20–22].

B1 — Arithmetic Mean (AM).

$$\hat{r}_{\text{tgt},k}^{\text{AM}} = \frac{1}{|\mathcal{N}'(t_k)|} \sum_{s_j \in \mathcal{N}'(t_k)} r_{j,k} \quad (16)$$

where $\mathcal{N}'(t_k) \subseteq \mathcal{N}(s_{\text{tgt}})$ are donors with valid records at t_k .

B2 — Linear Temporal Interpolation (LTI). Computes the donor mean series $\bar{r}_k = |\mathcal{N}'(t_k)|^{-1} \sum_{s_j} r_{j,k}$ and fills the missing segment by linear interpolation between the last pre-gap and first post-gap values [13].

B3 — Cubic Spline Interpolation (CSI). As B2 but using third-order polynomial spline interpolation; negative values are clipped to zero. Gaps exceeding 60 days are not filled [14].

B4 — Nearest-Neighbour Interpolation (NNI). Applies nearest-neighbour resampling to the donor mean series; same 60-day gap-length clip as B3 [14].

3.6 Evaluation Metrics

Reconstruction accuracy is assessed using four standard hydrometeorological metrics [13, 14]:

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{k=1}^n (\hat{r}_k - r_k^{\text{ref}})^2} \quad (17)$$

$$\text{MAE} = \frac{1}{n} \sum_{k=1}^n |\hat{r}_k - r_k^{\text{ref}}| \quad (18)$$

$$\text{NSE} = 1 - \frac{\sum_{k=1}^n (\hat{r}_k - r_k^{\text{ref}})^2}{\sum_{k=1}^n (r_k^{\text{ref}} - \overline{r^{\text{ref}}})^2} \quad (19)$$

$$\text{PBIAS} = 100 \times \frac{\sum_{k=1}^n (r_k^{\text{ref}} - \hat{r}_k)}{\sum_{k=1}^n r_k^{\text{ref}}} \quad (20)$$

where n is the total number of evaluated time steps; $\hat{r}_k$ and r_k^{ref} are the reconstructed and reference values; $\overline{r^{\text{ref}}}$ is their mean. Lower RMSE and MAE indicate better accuracy; $\text{NSE} = 1$ is perfect; $\text{PBIAS} = 0$ indicates no systematic bias (positive PBIAS = underestimation).

4 Results

4.1 Anomaly Detection

Supplementary Table S3 summarises the detection results across all 100 stations.

MDA carries the heaviest individual penalty (40–100 points per event) and is the primary driver of critical HS values. CFVA events, though fewer in number, involved prolonged durations (up to 134 consecutive days of stuck-sensor readings) that severely degrade data quality for affected stations. IPA and NTA events individually deduct only 5 points each, but their widespread occurrence across 29 and 24 stations, respectively, indicates minor but pervasive sensor noise.

4.2 Health Score Distribution

Supplementary Table S4 summarises the health score distribution.

Health scores span from -95 to 100 (mean $\pm$ SD = 75.2 ± 28.4). The spatial pattern (Figure 1a) reveals that low-HS stations tend to cluster in remote mountain sub-catchments, consistent with infrequent maintenance access.

4.3 Reconstruction Performance

A leave-one-out cross-validation experiment was conducted: each of the 62 high-quality stations was successively treated as a synthetic target, its records were replaced with NaN over gap patterns identical to those identified in actual low-HS stations, and all five methods (1DW and B1–B4) were applied using

the remaining stations as the donor pool. Reconstructed values were compared against the withheld true observations.

Figures 6 and 7 visualise the reconstruction results; Table 1 reports the quantitative comparison.

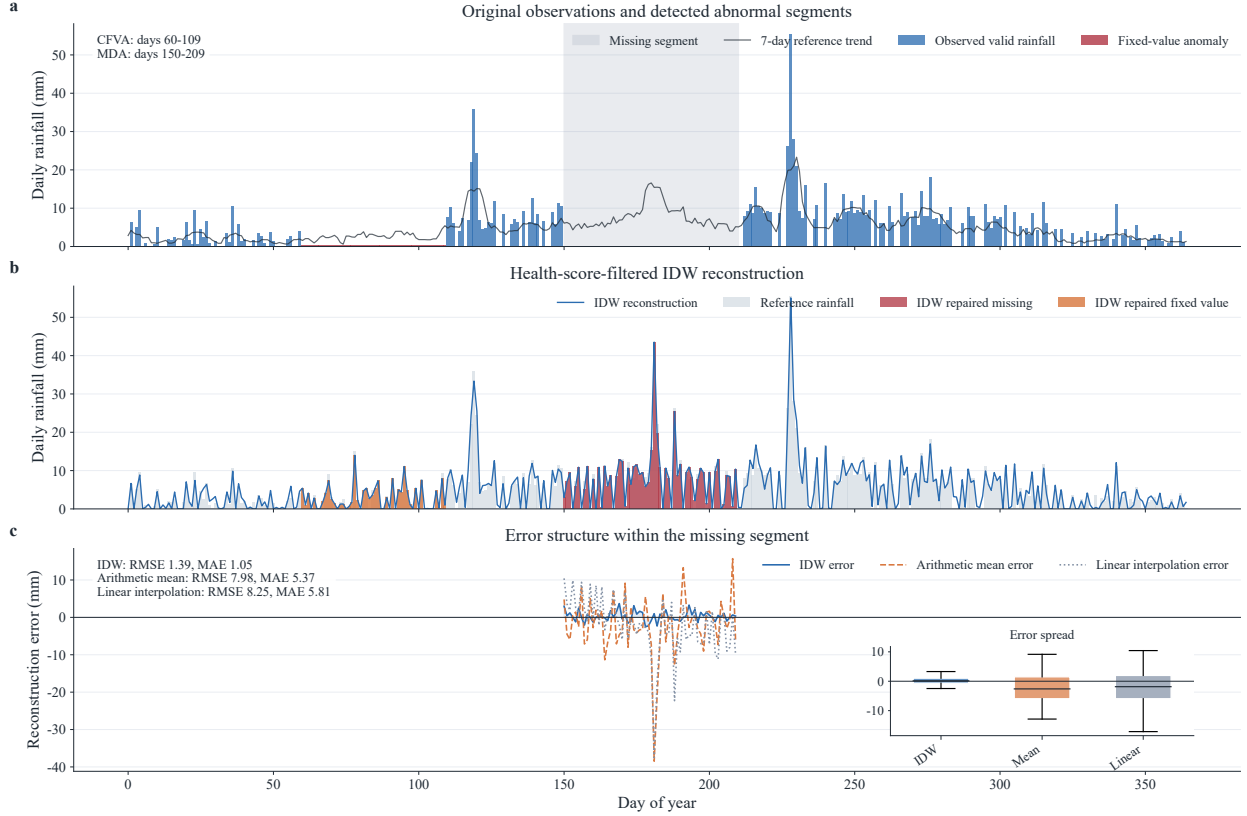


Figure 6: Rainfall reconstruction for a selected target station. **(a)** Original observations with detected anomalies highlighted. **(b)** IDW-reconstructed rainfall overlaid on the reference. **(c)** Reconstruction errors for IDW, arithmetic mean, and linear interpolation within the missing segment.

IDW’s RMSE (0.281 mm) is 58.9% lower than the next best method (CSI, 0.651 mm) and 90.8% lower than the worst (NNI, 3.065 mm). The NSE of 0.979 indicates near-perfect reconstruction skill. The negative NSE values for LTI and NNI reveal that temporal interpolation methods, when applied to extended gaps (> 45 days), produce estimates worse than simply using the long-term mean.

5 Discussion

5.1 Physical Motivation of Detection Thresholds

The CFVA detection set $\mathcal{V}_{\text{fixed}} = \{0.5, 1.0, 1.5\}$ mm is determined by the tipping-bucket resolution (0.5 mm/tip): stuck-sensor readings always land on exact multiples of the tip volume [6]. The run-length threshold $L_{\text{min}} = 50$ records was chosen empirically: it is large enough to exclude genuine light drizzle events but small enough to flag persistent mechanical failures. The 45-day MDA threshold reflects the Meishan Bureau’s maintenance schedule [17]: field teams inspect remote stations on a 45-day cycle, making this a natural boundary between recoverable short outages and confirmed equipment failures. This maintenance-cycle-aligned threshold design is broadly consistent with the tiered quality control protocols recommended for operational monitoring

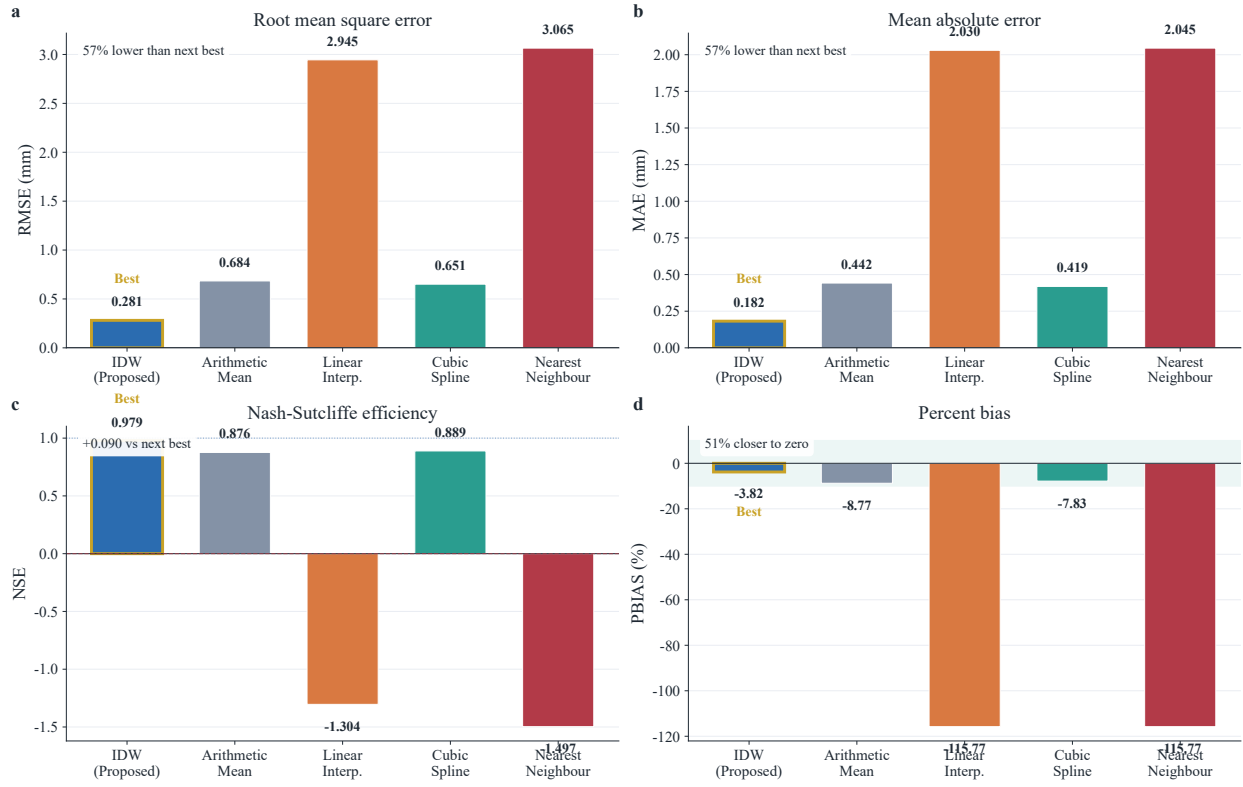


Figure 7: Performance comparison of five interpolation methods. Gold borders mark the best-performing method. (a) RMSE; (b) MAE; (c) NSE; (d) PBIAS.

Table 1: Quantitative comparison of five reconstruction methods (leave-one-out cross-validation). Bold denotes the best value in each column.

Method	RMSE (mm)	MAE (mm)	NSE	PBIAS (%)
IDW (Proposed)	0.281	0.182	0.979	-3.82
Arithmetic Mean (B1)	0.684	0.442	0.876	-8.77
Cubic Spline (B3)	0.651	0.419	0.889	-7.83
Linear Temporal (B2)	2.945	2.030	-1.304	-115.77
Nearest Neighbour (B4)	3.065	2.045	-1.497	-115.77

networks [3, 5].

5.2 Health Score as an Operational Tool

The HS framework converts complex, multi-dimensional anomaly information into a single actionable number. Stations with $HS < 0$ (9 in this network) require immediate field inspection and cannot contribute to spatial analyses. The decomposition of HS by anomaly type (Figure 4b) allows engineers to diagnose the probable cause: a large MDA component suggests communication or power failure; a dominant CFVA component points to bucket clogging. This diagnostic transparency is an advantage over purely data-driven quality indices that produce scores without interpretable components [3, 12]. The framework is analogous in spirit to the quality index (QI) approach in Osrodka et al. [3] but extends it from individual-measurement-level scoring to station-level aggregated diagnosis with operational maintenance prioritisation.

5.3 Why IDW Outperforms Temporal Methods

The superior performance of IDW over LTI and NNI reflects a fundamental difference in information source: IDW exploits **concurrent spatial observations** from neighbouring stations, whereas LTI and NNI rely on **temporal extrapolation** from sparse boundary values at the gap edges. For the Meishan network, where inter-storm dry periods can span weeks and monsoon fronts affect multiple stations simultaneously, concurrent spatial observations provide dramatically more relevant information. The advantage of IDW over the arithmetic mean (B1) derives from the spatial weighting: nearby donors receive proportionally higher weight, which is appropriate given the spatial correlation structure of monsoon rainfall (correlation length $\sim 15\text{--}25$ km for the Meishan region) [18, 23]. Recent comparison studies confirm that IDW remains competitive with kriging and machine-learning methods when inter-station distances are short and the donor pool is well-curated — exactly the conditions created by the health-score filter [15, 24].

5.4 Limitations and Future Work

Several limitations should be acknowledged. First, the fixed search radius $d_0 = 5$ km was calibrated for valley stations; future work should investigate adaptive radius selection based on network density and local terrain classification. Second, the current implementation does not account for topographic modulation of precipitation. Elevation-corrected IDW or kriging-with-external-drift formulations incorporating digital elevation model (DEM) data would be natural extensions [15, 25]. Third, the cross-validation protocol uses synthetic gap patterns derived from actual low-HS stations applied to high-HS stations; a true held-out validation on independently failed stations would provide additional evidence of generalisation. Fourth, real-time deployment would require incremental anomaly detection algorithms and pre-cached weight matrices to meet the sub-second latency requirements of operational flood forecasting systems [7].

6 Conclusion

This study presented a health-score-driven framework for the intelligent diagnosis and spatial reconstruction of rainfall monitoring data. The main conclusions are:

1. The four-category anomaly detection algorithm (CFVA, MDA, IPA, NTA) successfully identified quality problems in 38 of 100 stations, with distinct anomaly patterns corresponding to distinct failure modes.
2. The Health Score ($HS_i = 100 - \sum P(a)$) provides a transparent, interpretable, and actionable quality index. Nine stations scored below zero (critical); 62 qualified as high-quality donors.

3. idw with health-score-filtered neighbours achieved the best reconstruction accuracy (RMSE = 0.281 mm, NSE = 0.979), outperforming all four benchmarks by 58.9%–90.8% in RMSE.
4. The framework is model-agnostic and operationally deployable as a REST API service integrated with existing hydrological information management infrastructure [17].

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Funding

This research was funded by the Youth Science and Technology Innovation Program of Anhui and Huaihe River Institute of Hydraulic Research, grant number KY202503.

Acknowledgements

Not applicable.

Author contributions

B.L. conceived the study, designed the methodology, led the analysis and drafted the manuscript. M.W. contributed to software implementation, data curation and validation. X.Z. contributed to investigation, interpretation of results and manuscript revision. W.Z. contributed to validation, operational interpretation and manuscript revision. All authors reviewed and approved the final manuscript.

Data availability

The data that support the findings of this study are available from Anhui and Huaihe River Institute of Hydraulic Research, but restrictions apply to the availability of these data, which were used under institutional confidentiality agreements for the current study, and so are not publicly available. Data are however available from the authors upon reasonable request and with permission of Anhui and Huaihe River Institute of Hydraulic Research.

Code availability

The source code implementing the anomaly detection, health scoring, spatial reconstruction and synthetic demonstration workflow is publicly available at <https://github.com/ArthurCode-prod/HS-IDW/>. A code snapshot is also provided as supplementary material.

Additional information

Supplementary Tables S1–S5 and Supplementary Note S1 accompany this manuscript. The authors declare no competing interests.

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